Practice Questions

MTH633

Lectures (29 – 31) or Modules (114 – 126)

Q1: Every subgroup H of an abelian group G is normal.

Q2: The set of all inner automorphisms of G is a subgroup of Aut(G).

Q3: A factor group of a cyclic group is cyclic.

Q4: Prove that Aut $(\mathbb{Z}_n) \cong U_n$.

Q5: Let H, a subgroup of a group G be a normal subgroup of G, then $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.